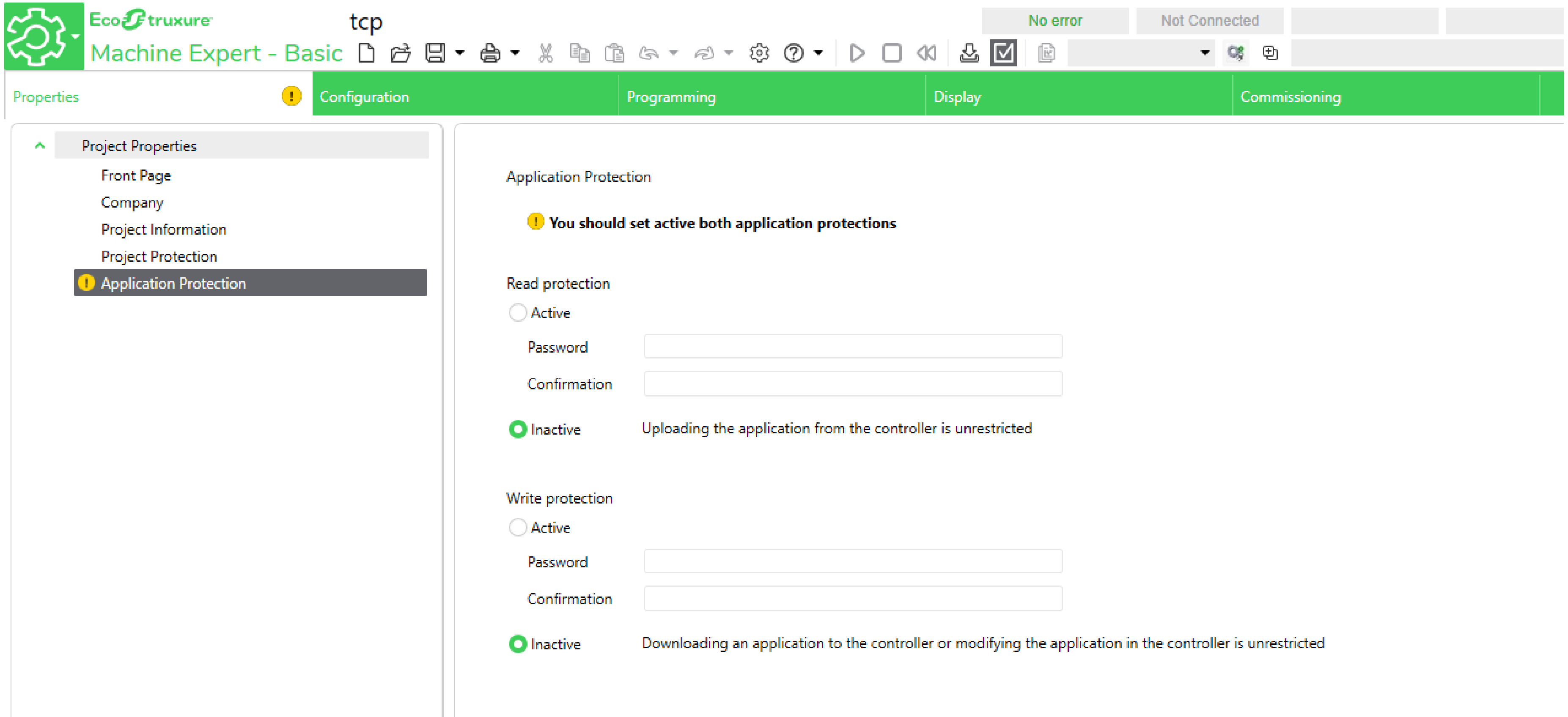
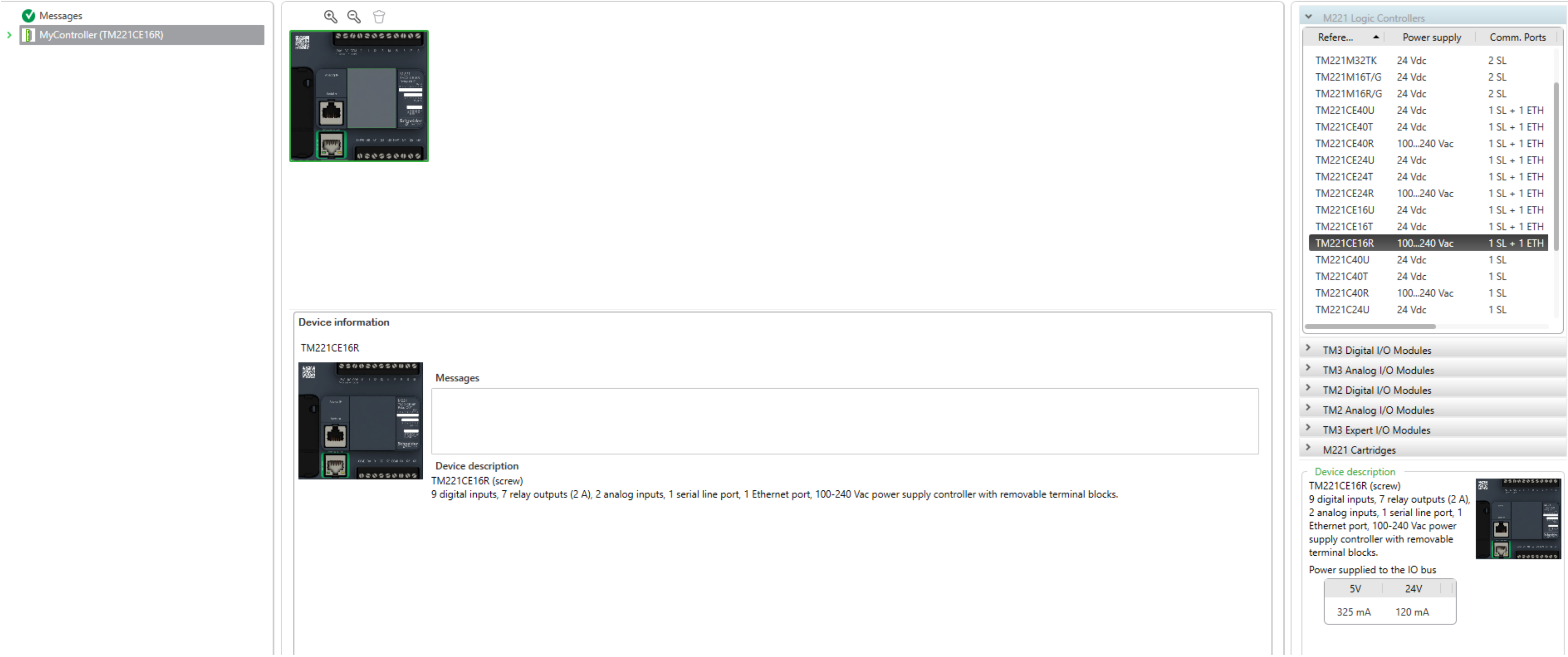
AI meet PLC Schneider

ROFIQ CAHYO PRAYOGO, S.T.,M.T.

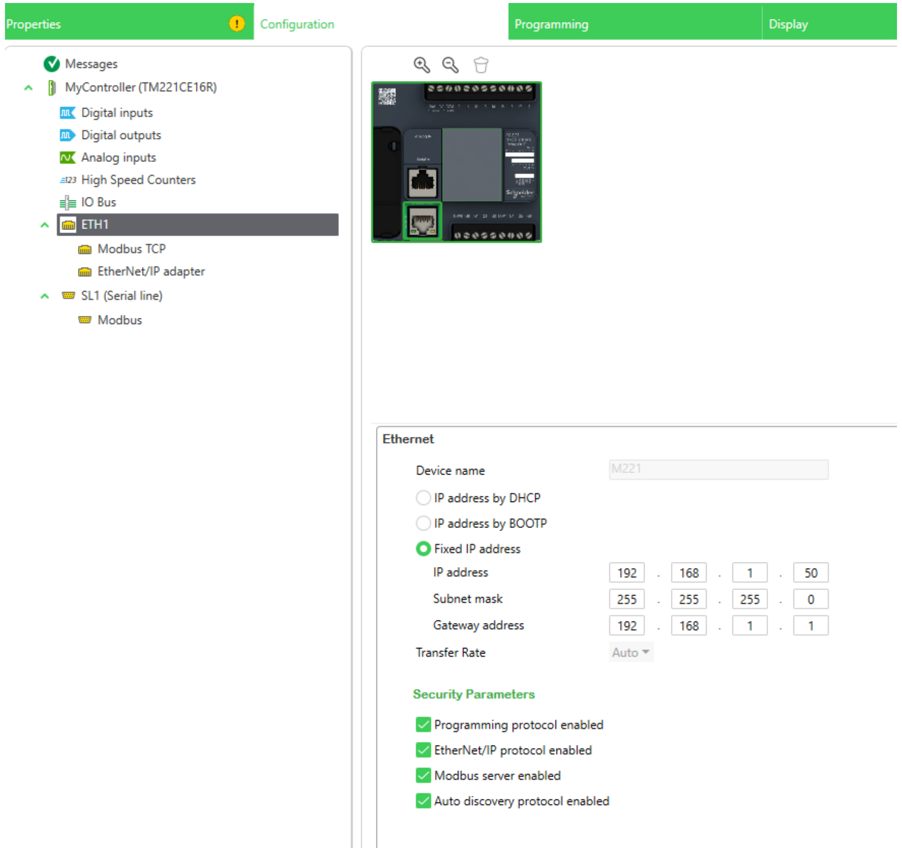
Config awal


EcoStruxure
Machine Expert - Basic
tcp
No error
Not Connected
Properties
Configuration
Programming
Display
Commissioning
Project Properties
Front Page
Company
Project Information
Project Protection
Application Protection
Application Protection
You should set active both application protections
Read protection
Active
Password
Confirmation
Inactive
Uploading the application from the controller is unrestricted
Write protection
Active
Password
Confirmation
Inactive
Downloading an application to the controller or modifying the application in the controller is unrestricted

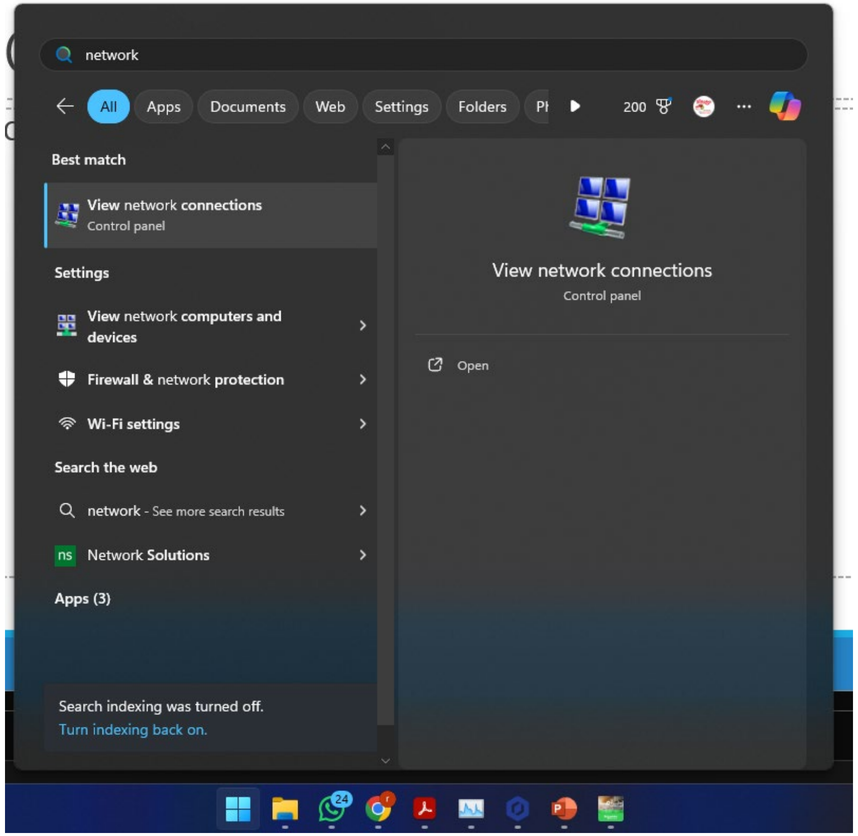
Pilih PLC TM221CE16R

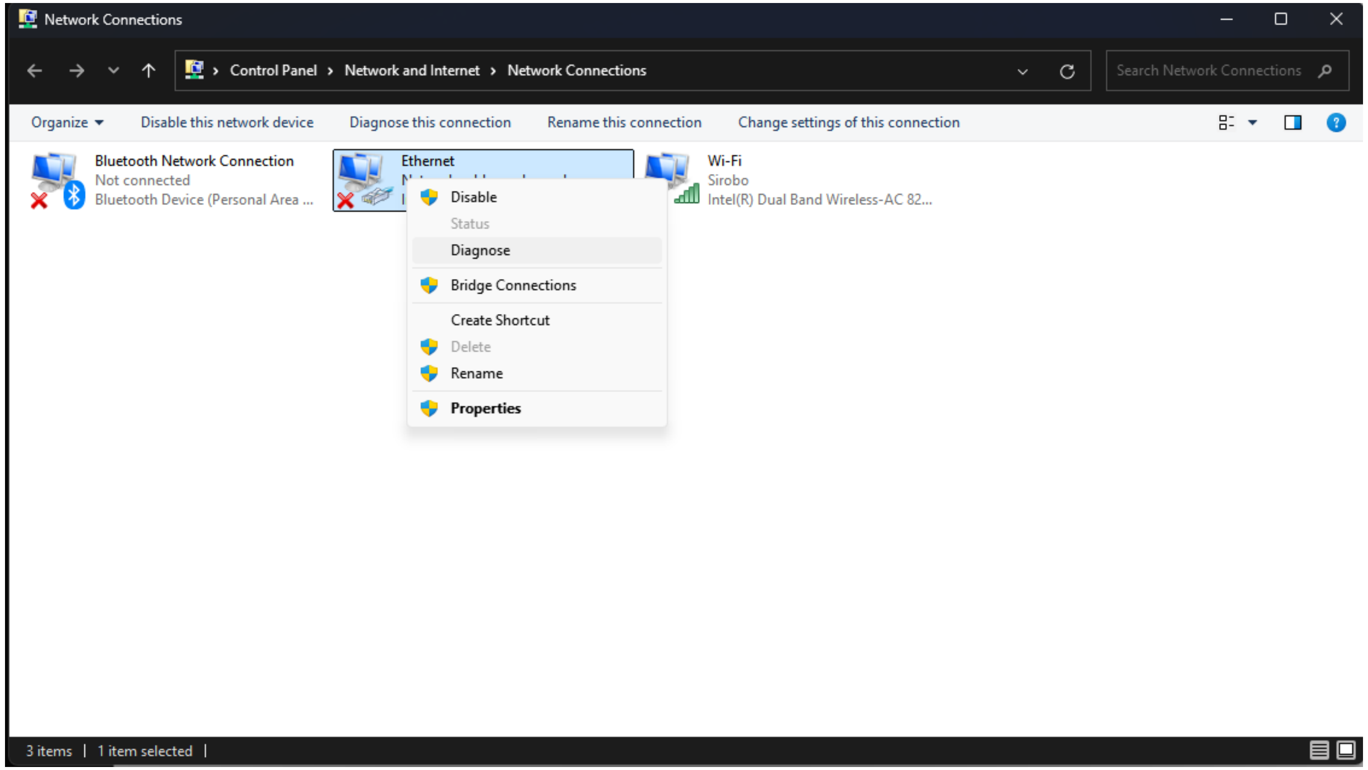


Setting Alamat IP di PLC 192.168.1.50

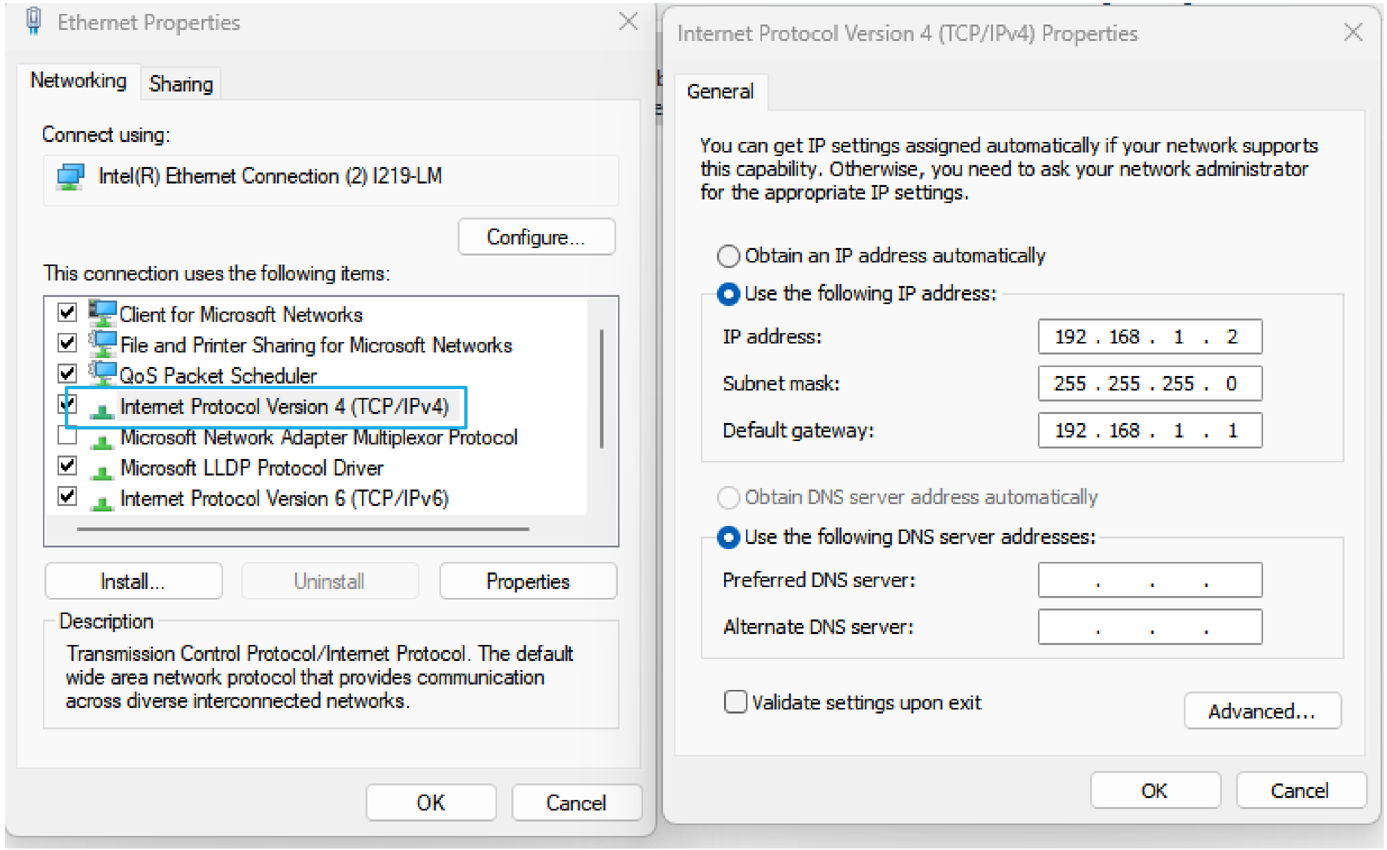


Setting Alamat IP di PC

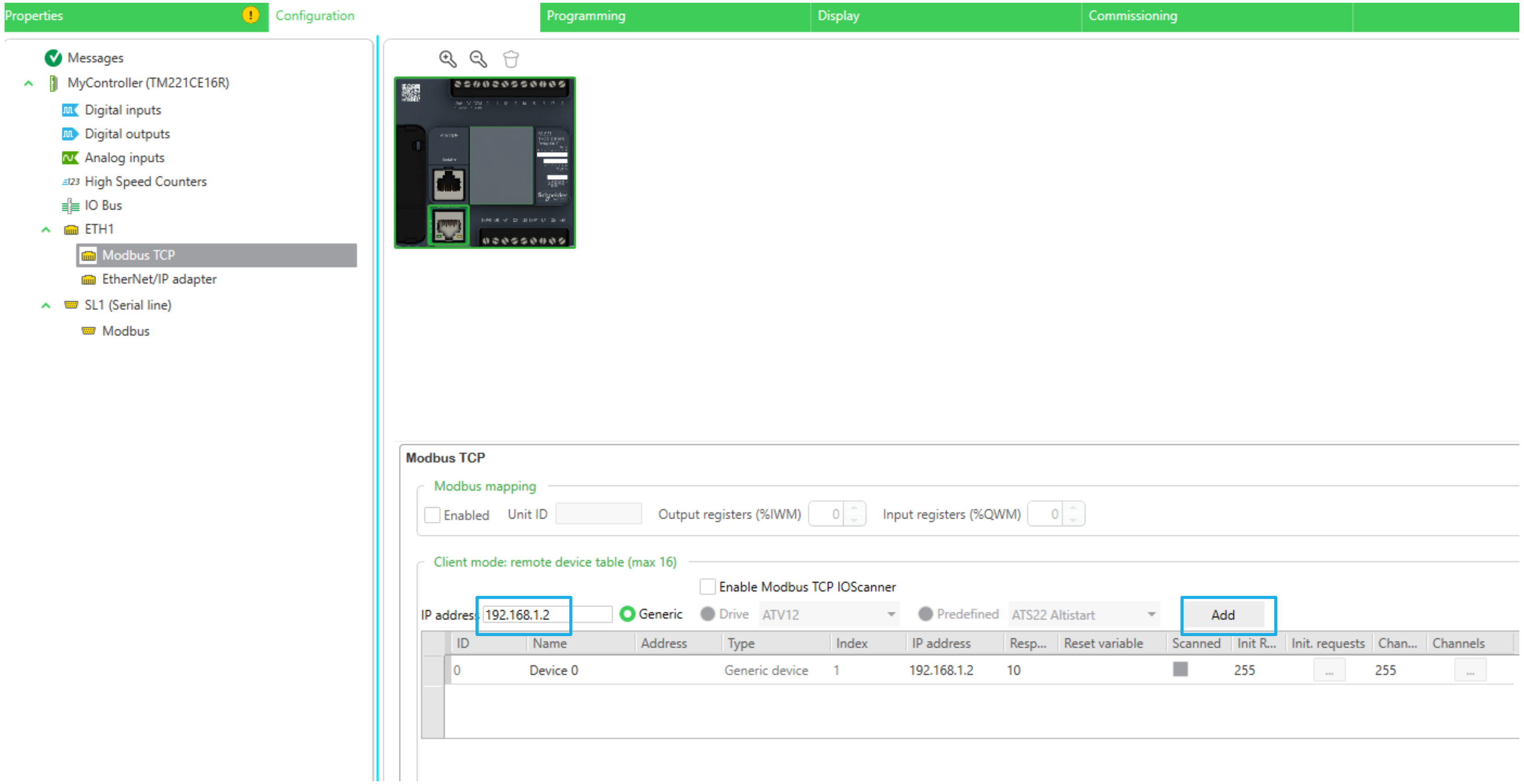




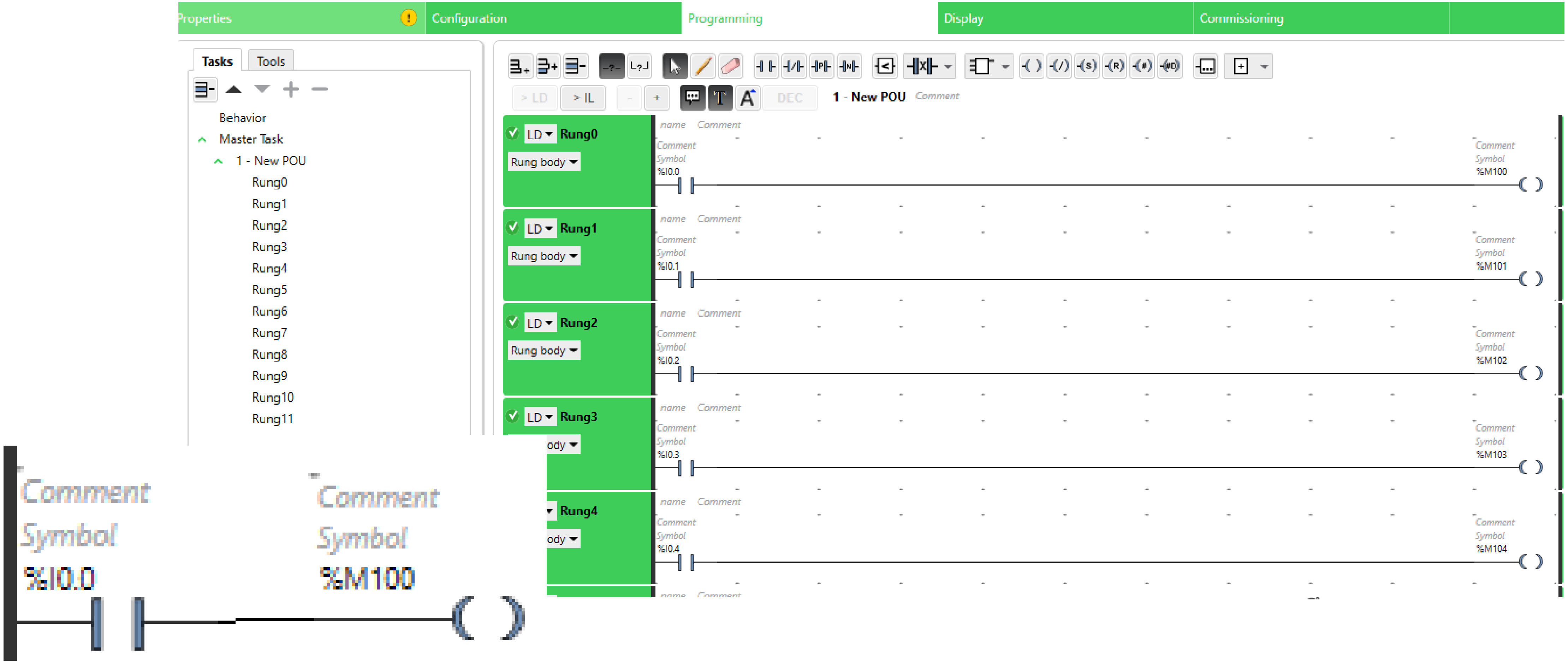
Setting Alamat IP di PC 192.168.1.2



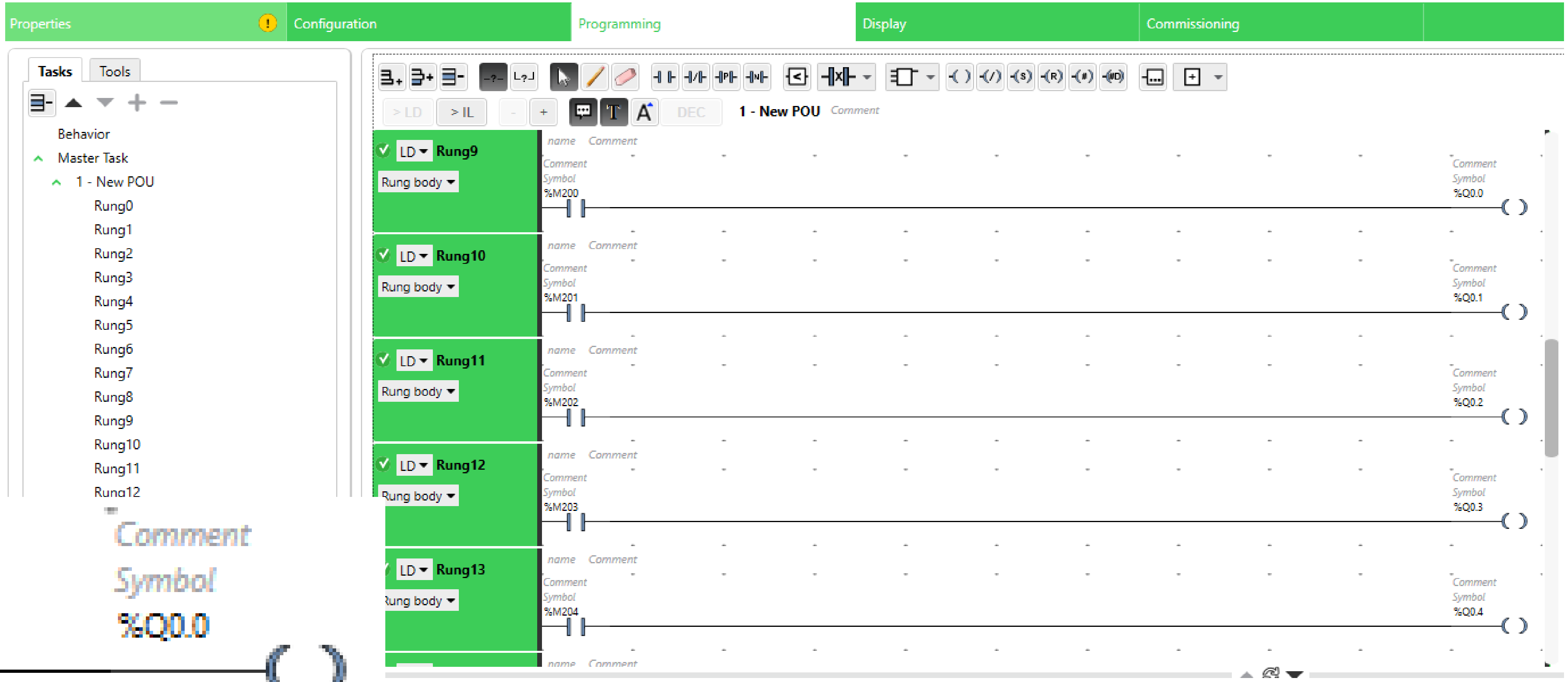
Setting Alamat IP dari PLC ke PC 192.168.1.2

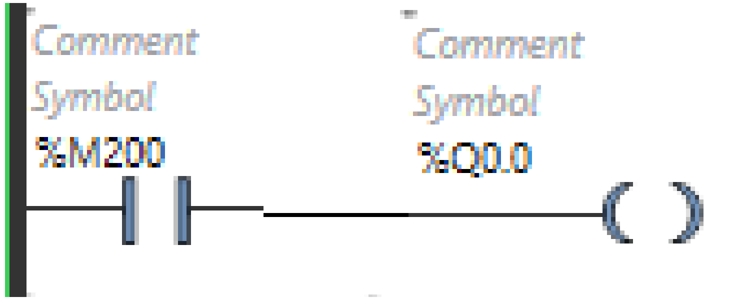


Baca I0 - I3 sambungkan ke M100 - M103

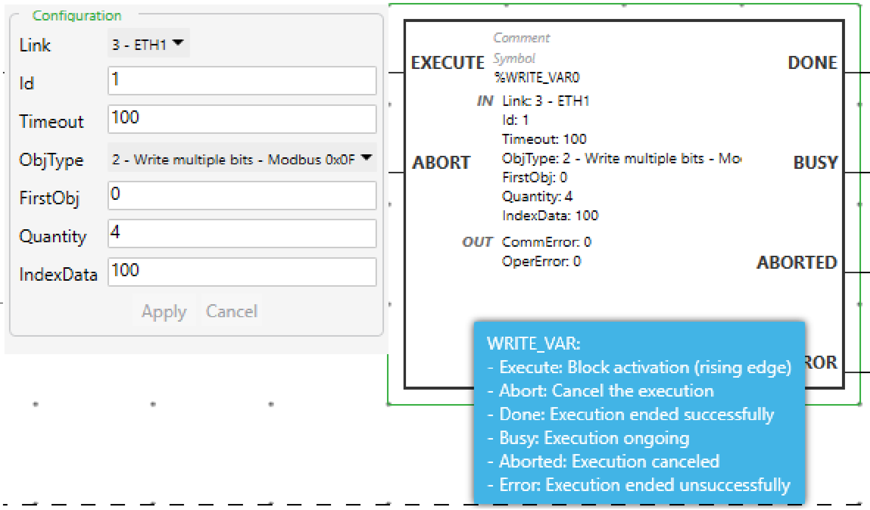


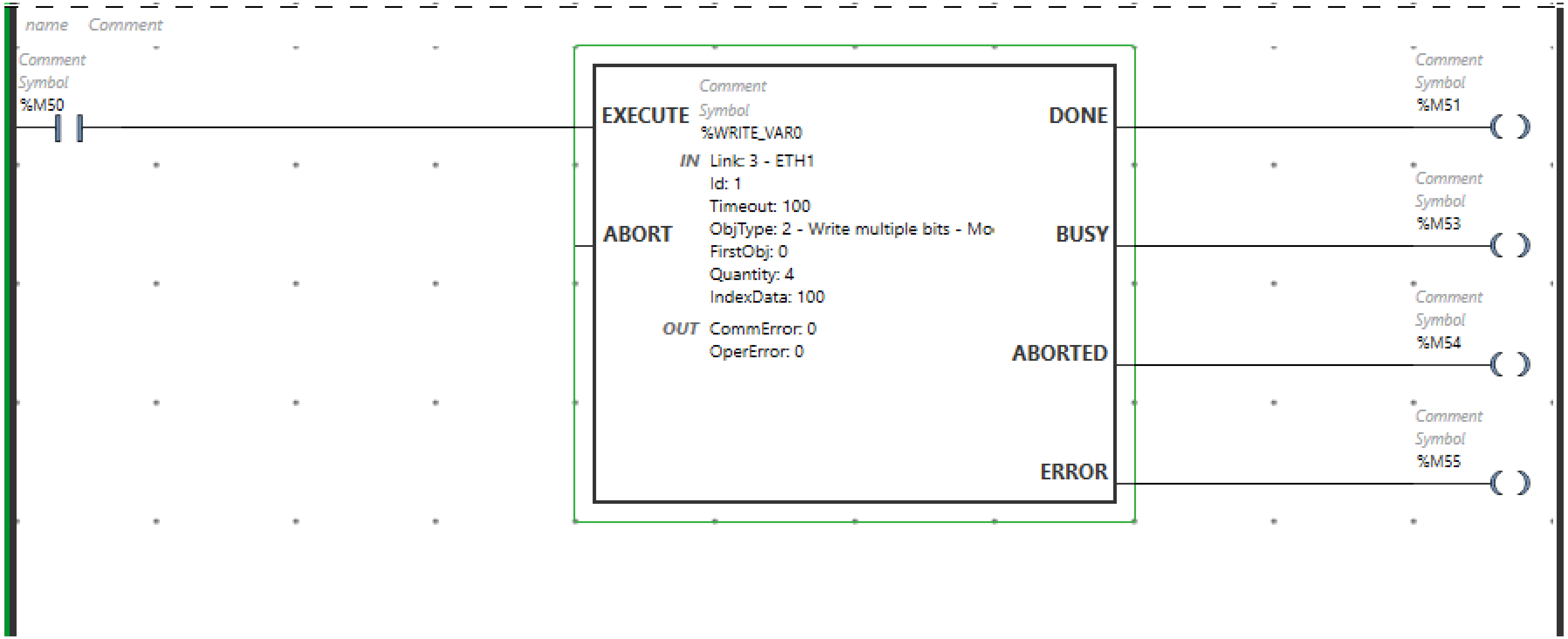
Baca M200 - M203 sambungkan ke Q0 - Q03

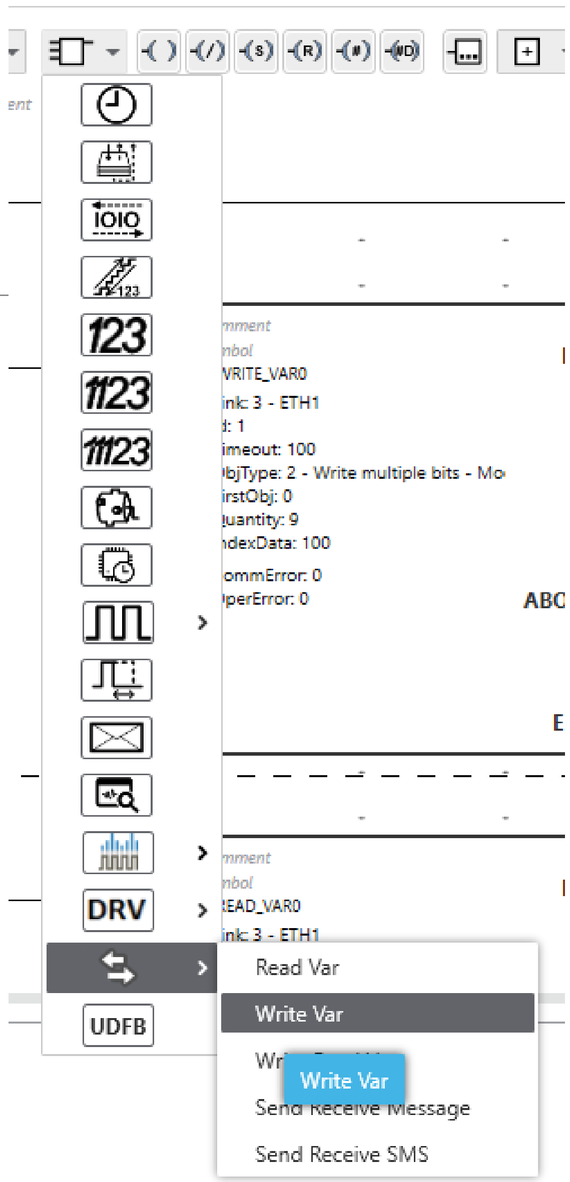




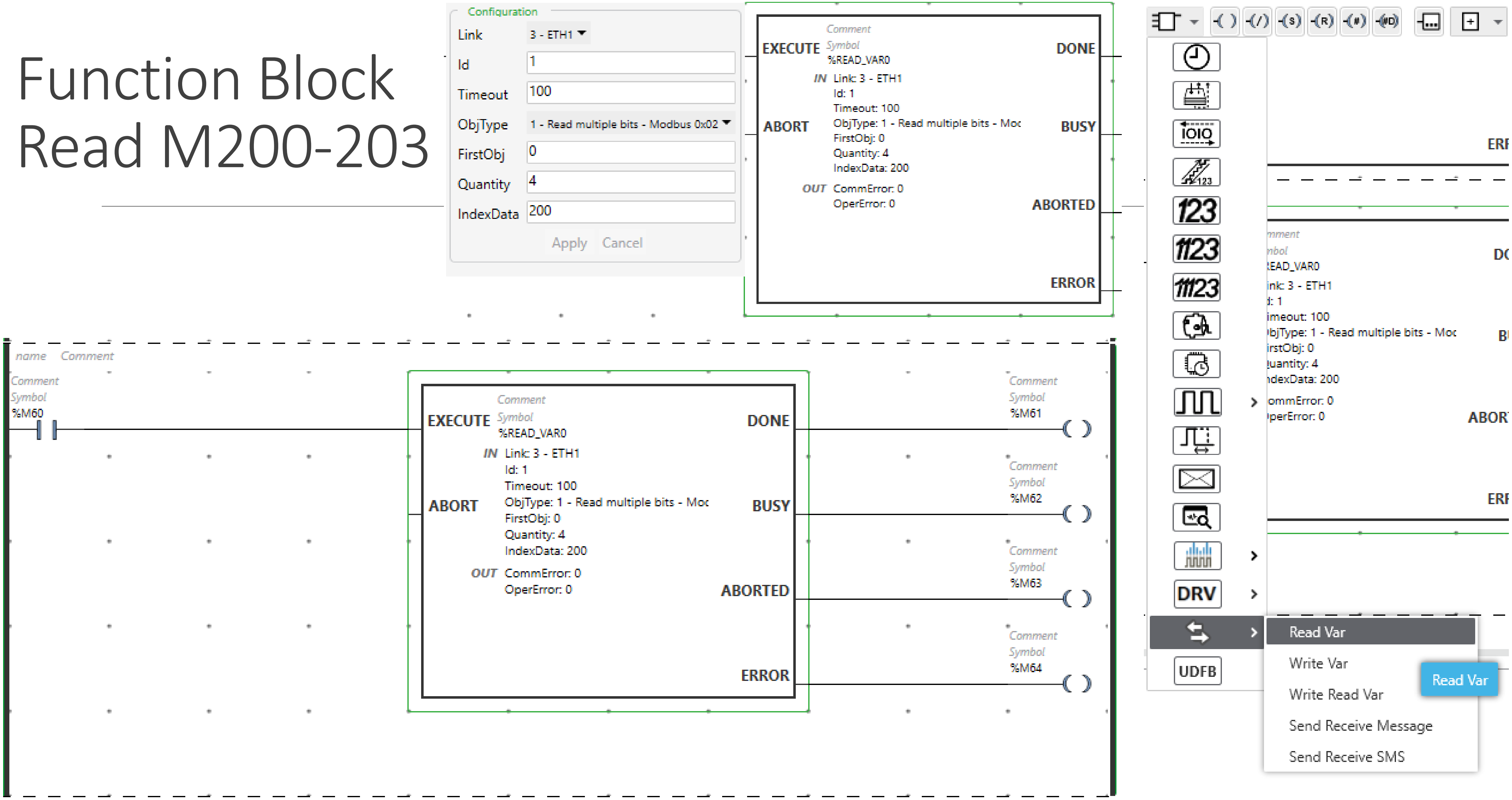
Function Block
Write M100-103



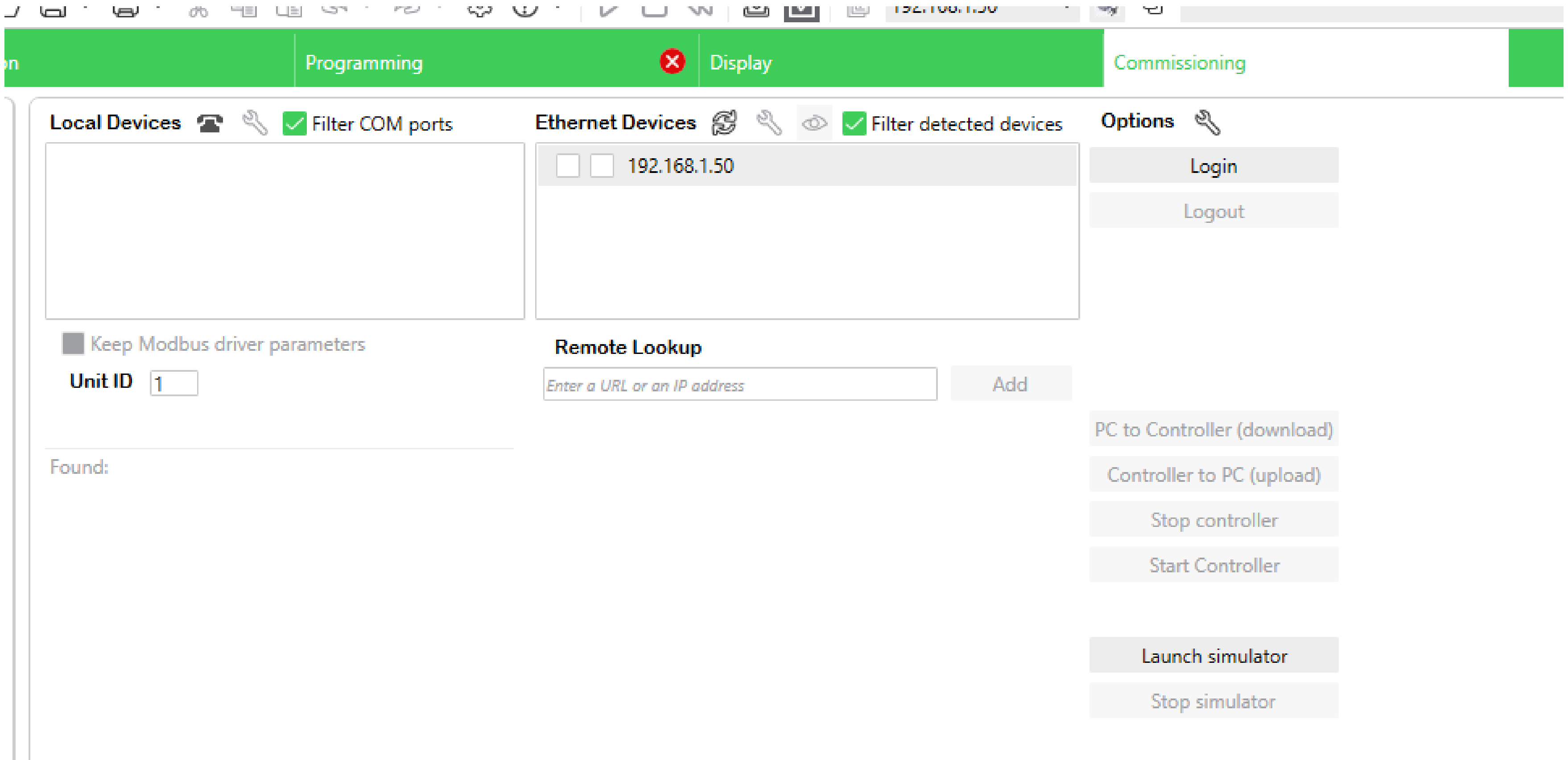




Function Block Read M200-203



Upload Program ke PLC via USB
Login - Download - Start



Program Python tes.py
untuk koneksi modbus tcp

https://chatgpt.com/share/68261b22-3b84-800f-9b37-f3d77403f6fa

Buatkan program python

Connect ke PLC via Modbus TCP

alamat PLC 192.168.1.50

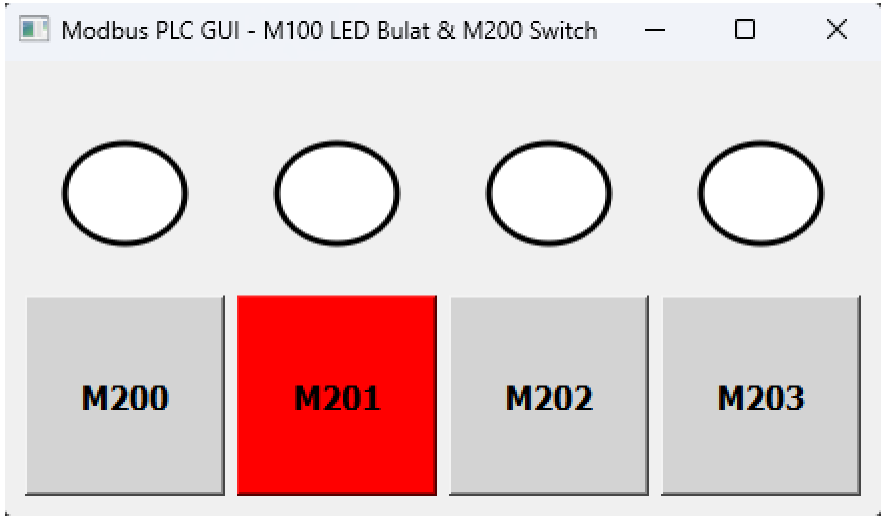
alamat PC 192.168.1.2

baca 4 data M100-M103 buatkan seperti LED

kirim 4 data M200-203 buatkan seperti Switch

buatkan GUI dengan QT

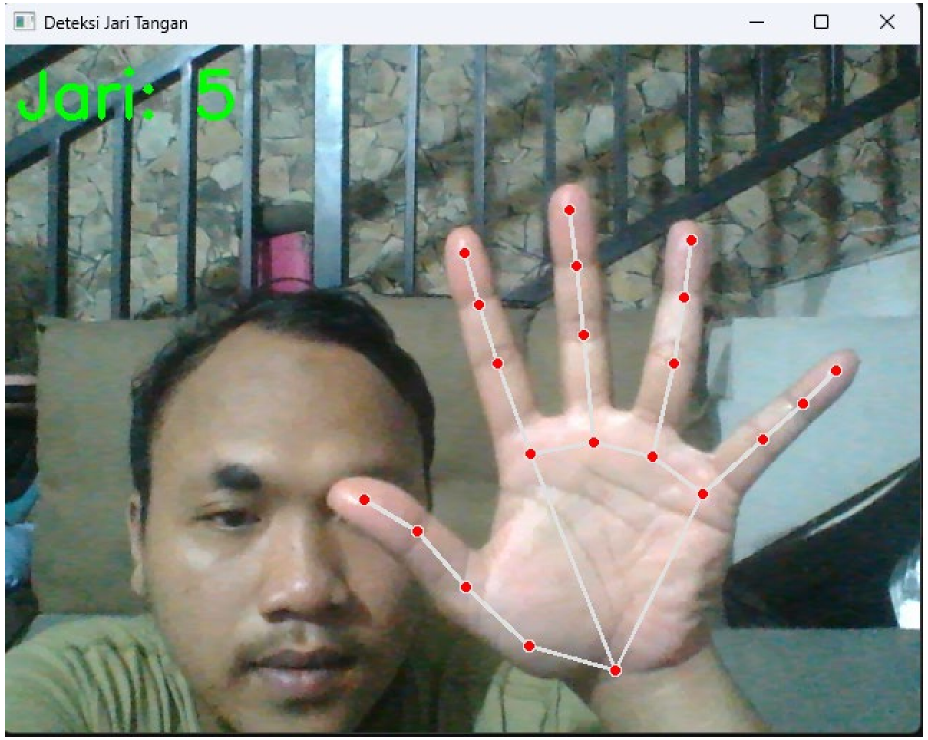
tampilkan LOG data di console



Program Python tes2.py untuk deteksi jari tangan

https://chatgpt.com/share/68261d76-ee1c-800f-b573-645bda348319

buatkan program python untuk deteksi jari tangan



Program Python tes3.py untuk gabungkan deteksi jari tangan dan koneksi modbus tcp

https://chatgpt.com/share/68262721-a978-800f-b4d0-2515e64697b5

Copy Paste program tes modbus tcp, shift + enter

Copy Paste program deteksi jari, shift + enter

gabungkan kedua program diatas,

ketika jari telunjuk diangkat M200 on

ketika jari tengah diangkat M201 on

ketika jari manis diangkat M202 on

ketika jari kelingking diangkat M203 on

Ketika mengepal atau turun, akan off

Ketika tidak terdeteksi jari, toggle tetap bisa di on off kan

tampilkan webcam di gui dan bawahnya gui seelumnya

Tugas

Video semua kegiatan ketika mencoba PLC

Buat Video dengan screenrecorder layar dan webcam

Jelaskan program PLC dari config hingga upload, serta config IP PC

Jelaskan program python tes koneksi modbus tcp

Jelaskan program python deteksi jari tangan

Jelaskan program python yang sudah terintergrasi

Upload di youtube kalian